

# Hadoop. Amazon Web Services

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Scalable Database CS673  
Assignment



# TABLE OF CONTENT

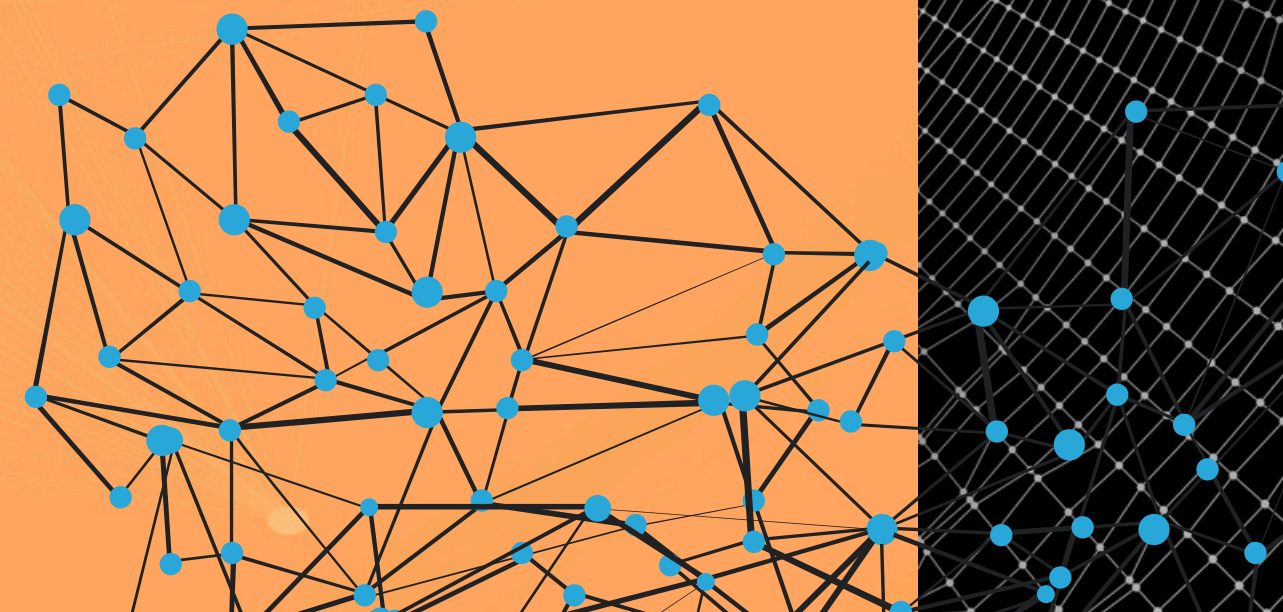
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## Scalable Database

- Creation EC2 Key Pair
- Creation the EMR cluster
- Setting up a connection to Hadoop

Upload readable screenshots of the all the steps of the Assignments as ONE single Word document.

Make sure you terminate the EMR cluster once the assignment is finished else the EC2 instances would run at the backend increasing the cost





# Creation EC2 Key Pair

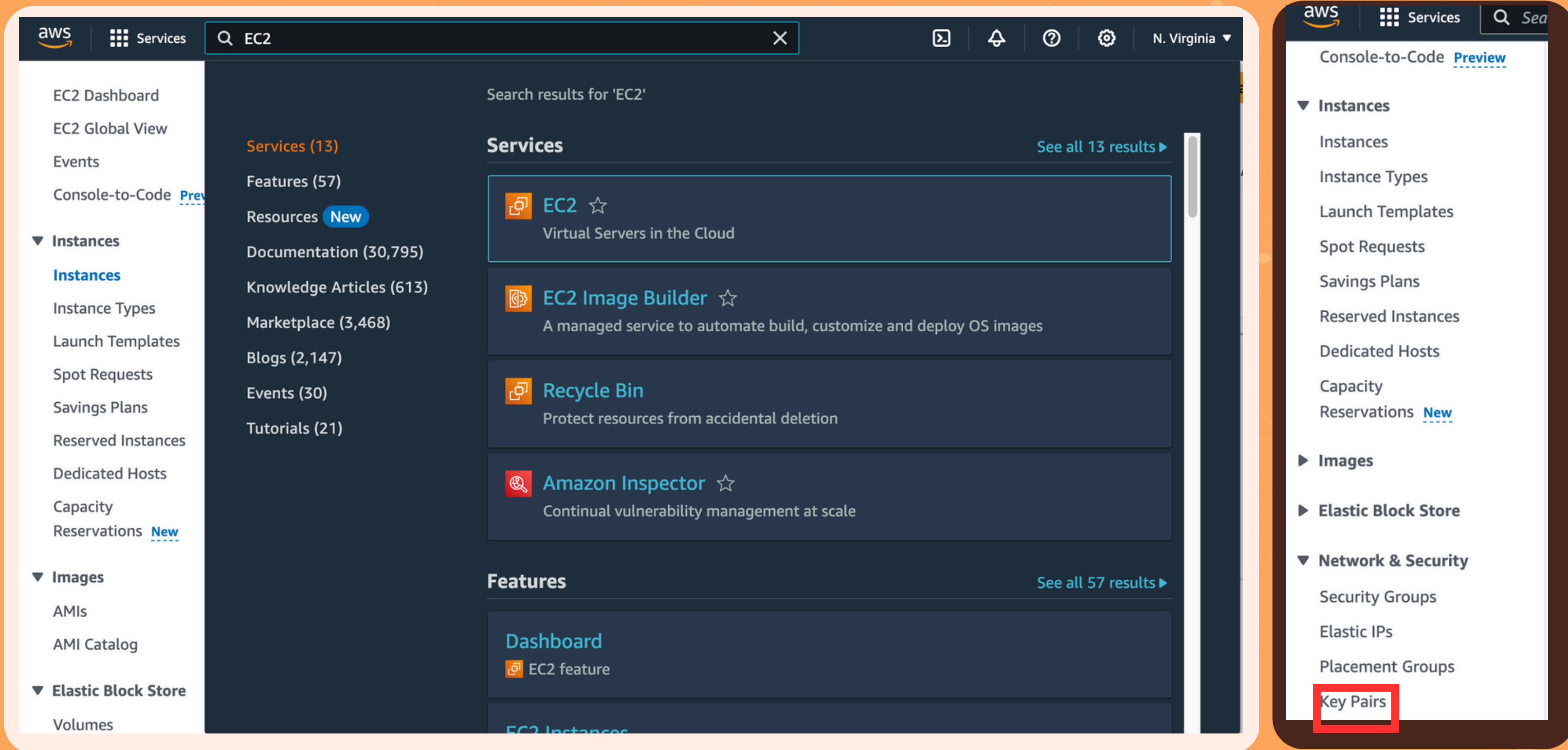




# 1. Creation an “EC2 Key Pair” to connect to the EMR cluster via any SSH client

a. Open the Amazon console and type “EC2” in the search bar of the dashboard, select it.

b. In the navigation pane, under Network & Security, choose Key Pairs as shown below:

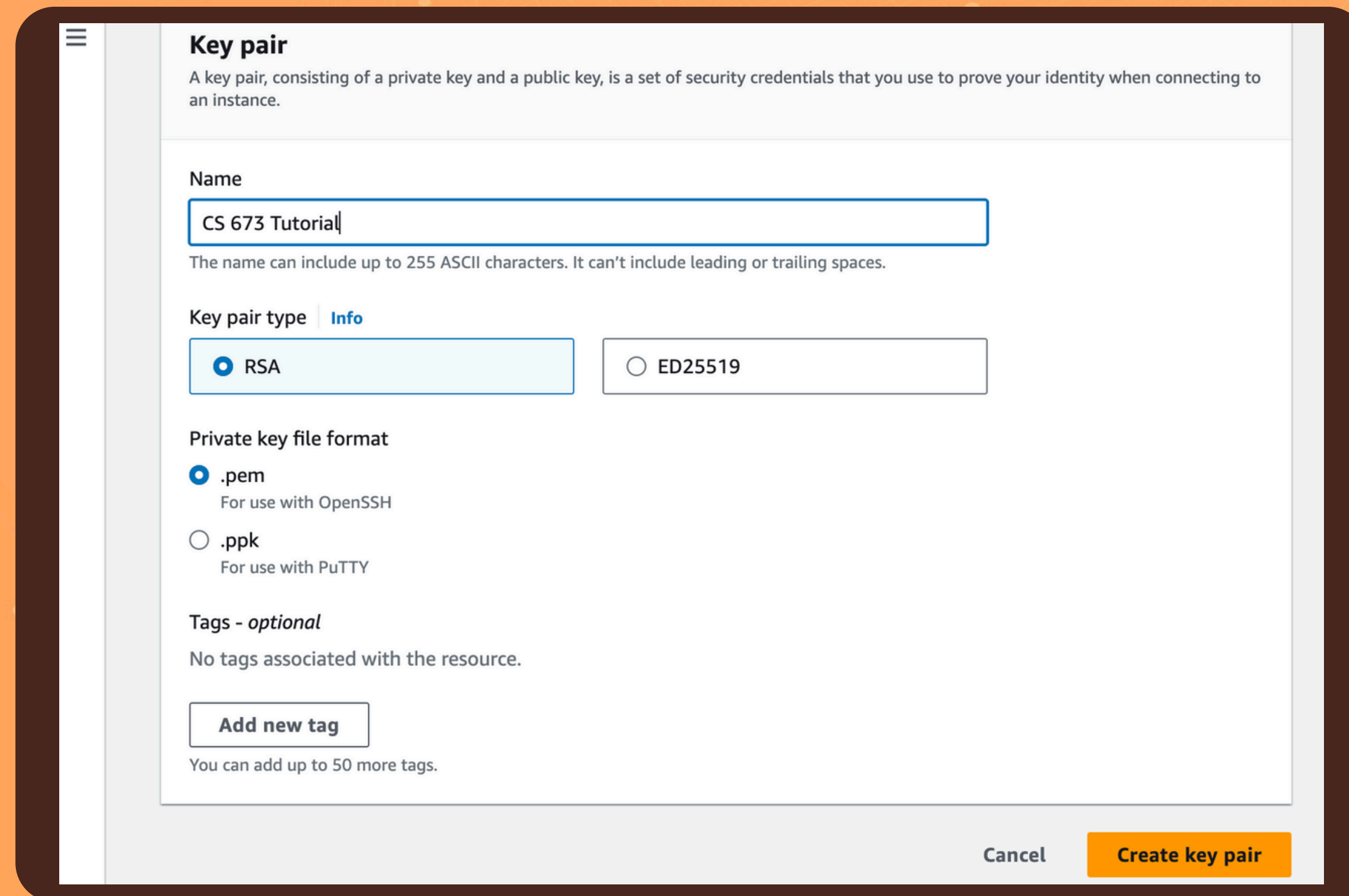




c. Choose **Create key pair**.

For Name, **enter** a descriptive name for the key pair. Amazon EC2 associates the public key with the name that you specify as the key name.

“**pem**” configuration is more suitable for Mac/Linux, “**ppk**” - for Windows Users.



The screenshot shows the 'Key pair' creation page in the AWS Management Console. The page title is 'Key pair' with a subtitle explaining that a key pair consists of a private key and a public key used for authentication. The 'Name' field contains 'CS 673 Tutorial' with a note that names can be up to 255 ASCII characters. Under 'Key pair type', the 'RSA' option is selected. Under 'Private key file format', the '.pem' option is selected, noted as being for use with OpenSSH. The '.ppk' option is also available, noted as being for use with PuTTY. There is a section for 'Tags - optional' with a note that no tags are currently associated and a button to 'Add new tag'. At the bottom right are 'Cancel' and 'Create key pair' buttons.

**Key pair**  
A key pair, consisting of a private key and a public key, is a set of security credentials that you use to prove your identity when connecting to an instance.

Name  
CS 673 Tutorial  
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type [Info](#)  
☒ RSA ☐ ED25519

Private key file format  
☒ .pem  
For use with OpenSSH  
☐ .ppk  
For use with PuTTY

Tags - *optional*  
No tags associated with the resource.  
[Add new tag](#)  
You can add up to 50 more tags.

Cancel **Create key pair**

d. For Key pair type, choose either **RSA**

e. The private key file is automatically downloaded by your browser. The base file name is the name that you specified as the name of your key pair, and the file name extension is determined by the file format that you chose. Save the private key file in a safe place.



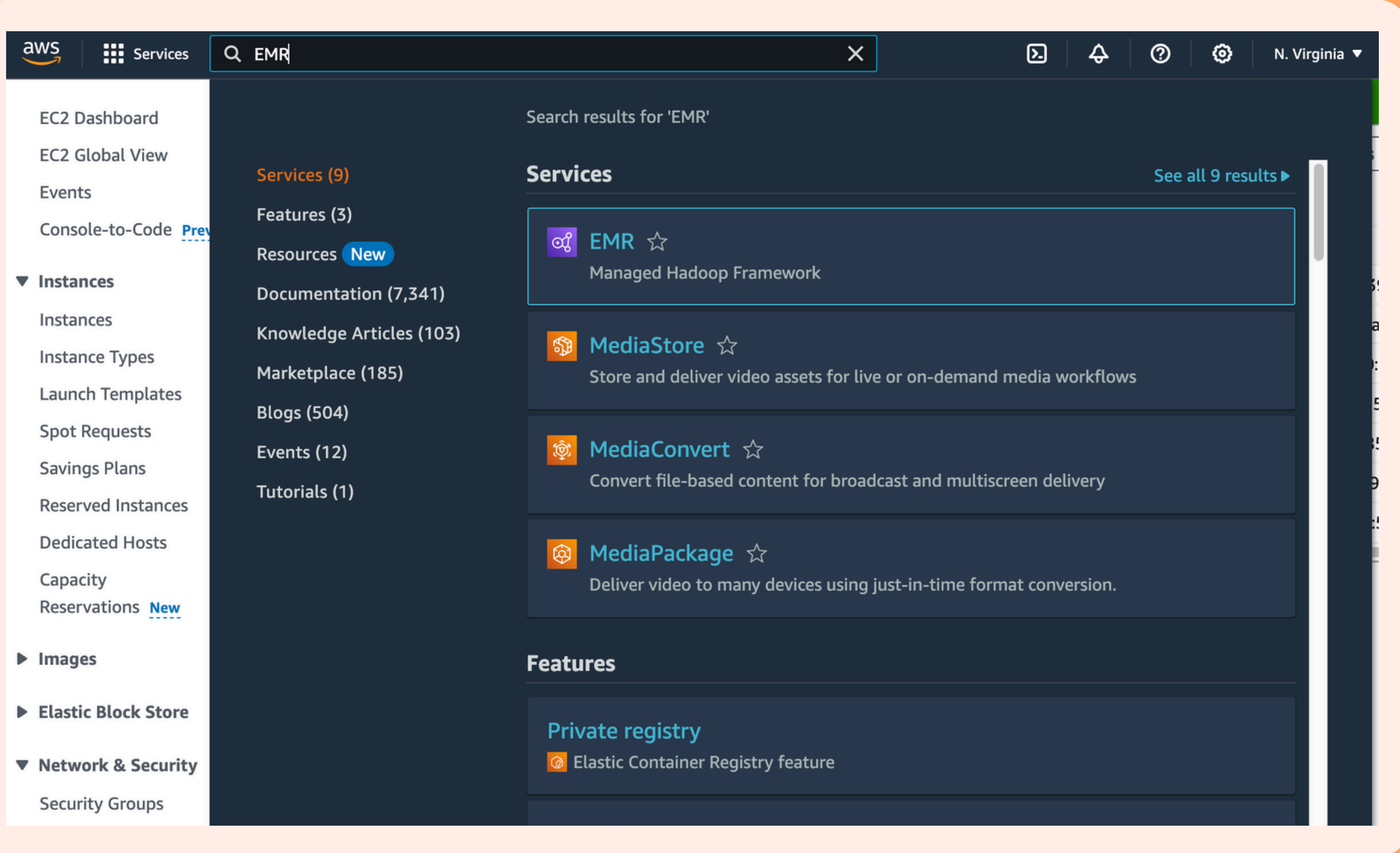
# Creation the EMR cluster





2. Creation the EMR cluster. First of all, please change youe region to “Ohio”.

a. Open the Amazon console and type “EC2” in the search bar of the dashboard, select it.





## b. Choose **Create cluster**.

- Provide a Name to the cluster and select any Release of EMR (Ex: emr-7.0.0)
- Keep the instance type as m5.xlarge and you can change the “Number of Instances” as per your requirement. It will be sufficient 1 instance for us.

aws Services Search [Option+S]

Name  
My\_First\_Cluster

Amazon EMR release [Info](#)  
A release contains a set of applications which can be installed on your cluster.  
emr-7.0.0

Application bundle

| Spark Interactive | Core Hadoop | Flink | HBase | Presto | Trino | Custom |
|-------------------|-------------|-------|-------|--------|-------|--------|
|                   |             |       |       |        |       |        |

|                                                           |                                                         |                                                |
|-----------------------------------------------------------|---------------------------------------------------------|------------------------------------------------|
| <input type="checkbox"/> AmazonCloudWatchAgent 1.300031.1 | <input type="checkbox"/> Flink 1.18.0                   | <input type="checkbox"/> HBase 2.4.17          |
| <input type="checkbox"/> HCatalog 3.1.3                   | <input checked="" type="checkbox"/> Hadoop 3.3.6        | <input checked="" type="checkbox"/> Hive 3.1.3 |
| <input checked="" type="checkbox"/> Hue 4.11.0            | <input type="checkbox"/> JupyterEnterpriseGateway 2.6.0 | <input type="checkbox"/> JupyterHub 1.5.0      |
| <input type="checkbox"/> Livy 0.7.1                       | <input type="checkbox"/> MXNet 1.9.1                    | <input type="checkbox"/> Oozie 5.2.1           |
| <input type="checkbox"/> Phoenix 5.1.3                    | <input checked="" type="checkbox"/> Pig 0.17.0          | <input type="checkbox"/> Presto 0.283          |
| <input type="checkbox"/> Spark 3.5.0                      | <input type="checkbox"/> Sqoop 1.4.7                    | <input type="checkbox"/> TensorFlow 2.11.0     |
| <input checked="" type="checkbox"/> Tez 0.10.2            | <input type="checkbox"/> Trino 426                      | <input type="checkbox"/> Zeppelin 0.10.1       |
| <input type="checkbox"/> ZooKeeper 3.5.10                 |                                                         |                                                |



- ▼

Cluster configuration - required

Info

Choose a configuration method for the primary, core and task node groups for your cluster.

☒ Instance groups

Choose one instance type per node group

☐ Instance fleets

Choose any combination of instance types within each node group

Instance groups

Primary

Choose EC2 instance type

m5.xlarge  
4 vCore   16 GiB memory   EBS only storage  
On-demand price: USD 0.192 per instance/hour  
Lowest spot price: \$0.070 (us-east-2c)

Actions ▼

☐ Use high availability  
Launch highly available, more resilient cluster with three primary nodes on On-Demand Instances. This configuration applies for the lifetime of your cluster. [Find out more](#)

► Node configuration - optional

Core

Remove instance group

Choose EC2 instance type

m5.xlarge  
4 vCore   16 GiB memory   EBS only storage  
On-demand price: USD 0.192 per instance/hour  
Lowest spot price: \$0.070 (us-east-2c)

Actions ▼

► Node configuration - optional



- Within “EC2 key pair” field; select the key pair which you created in Step 1.

Override the default configurations for specific applications on your cluster.

### ▼ Security configuration and EC2 key pair [Info](#)

Choose a security configuration or create a new one that you can reuse with other clusters.

#### Security configuration

Select your cluster encryption, authentication, authorisation and instance metadata service settings.

#### Amazon EC2 key pair for SSH to the cluster [Info](#)

### ▼ Identity and Access Management (IAM) roles - required [Info](#)

Choose or create a service role and instance profile for the EC2 instances in your cluster.

#### Amazon EMR service role [Info](#)

The service role is an IAM role that Amazon EMR assumes to provision resources and perform service-level actions with other AWS services.

☒ Choose an existing service role  
Select a default service role or a custom role with IAM policies attached so that your cluster can interact with other AWS services.

☐ Create a service role  
Let Amazon EMR create a new service role so that you can grant and restrict access to resources in other AWS services.

Service role

#### EC2 instance profile for Amazon EMR

The instance profile assigns a role to every EC2 instance in a cluster. The instance profile must specify a role that can access the resources for your steps and bootstrap actions.

☒ Choose an existing instance profile  
Select a default role or a customised instance profile with IAM policies attached so that your cluster can interact with your resources in Amazon S3.

☐ Create an instance profile  
Let Amazon EMR create a new instance profile so that you can specify a customised set of resources for it to access in Amazon S3.

Instance profile

- Select Service role - “EMR\_DefaultRole”
- Select Instance profile - “DemoRoleForEC2”
- If you do not have “Default and Demo Roles”, please click on “Create a service role” and on “Create an instance profile” by keeping all default configurations.

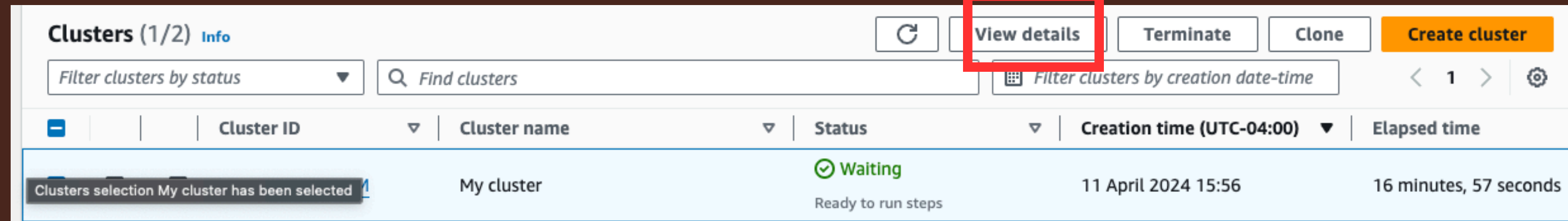
After clicking on Create Cluster, it will take around 10-15 minutes for the cluster to go in running state. In the meantime, follow step 3





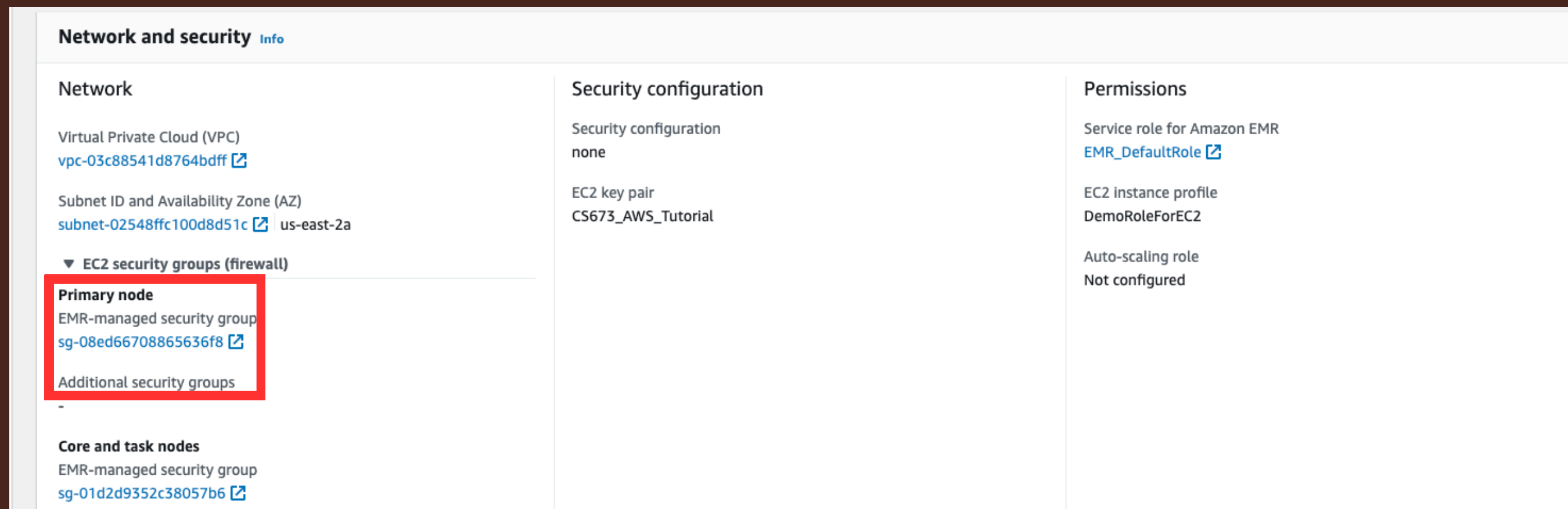
c. Now we need to add an inbound rule to the security group of Master Node which would allow all inbound connections on Port 22.

- Go to EMR and click on “Clusters”
- Click on the view details of the cluster which you have created.



The screenshot shows the AWS EMR Clusters console. At the top, there are buttons for 'View details' (highlighted with a red box), 'Terminate', 'Clone', and 'Create cluster'. Below these are search and filter options. A table lists the clusters, with one cluster named 'My cluster' in a 'Waiting' state, ready to run steps, created on 11 April 2024 at 15:56, and having elapsed 16 minutes and 57 seconds. A notification at the bottom left states 'Clusters selection My cluster has been selected'.

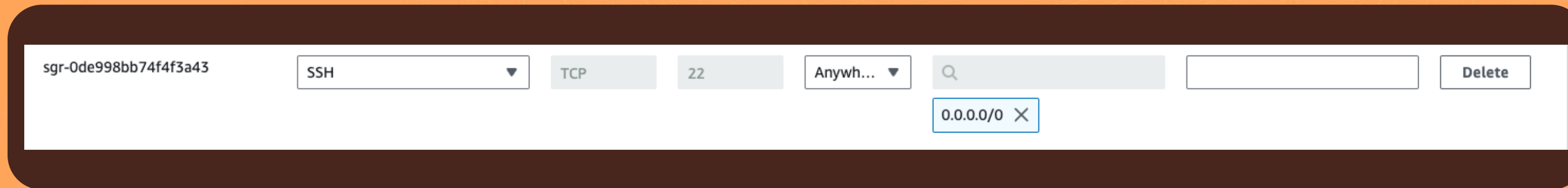
- Scroll down and check the details within “Network and security”
- Click on the “Security group id” for Primary node, it will lead you to ElasticMapReduce master



The screenshot shows the 'Network and security' tab for an EMR cluster. It is divided into three sections: Network, Security configuration, and Permissions. The 'Network' section shows the VPC ID (vpc-03c88541d8764bdf) and Subnet ID (subnet-02548ffc100d8d51c) in us-east-2a. Under 'EC2 security groups (firewall)', the 'Primary node' section is highlighted with a red box, showing the EMR-managed security group ID 'sg-08ed66708865636f8'. The 'Core and task nodes' section shows the EMR-managed security group ID 'sg-01d2d9352c38057b6'. The 'Security configuration' section shows 'Security configuration none' and 'EC2 key pair CS673\_AWS\_Tutorial'. The 'Permissions' section shows the 'Service role for Amazon EMR' as 'EMR\_DefaultRole', the 'EC2 instance profile' as 'DemoRoleForEC2', and the 'Auto-scaling role' as 'Not configured'.



- Scroll Down and click on “Edit inbound rules” -> Scroll down and click on “Add rule” -> Select Type as “SSH” and Source as “Anywhere-IPv4” as shown below:

A screenshot of a web interface for configuring a rule. The interface is dark-themed with a light-colored bar at the top. On the left, there is a text input field containing the identifier 'sgr-0de998bb74f4f3a43'. To its right is a dropdown menu showing 'SSH'. Further right is a button labeled 'TCP'. Next is a text input field containing '22'. To its right is another dropdown menu showing 'Anywh...'. Below this dropdown is a search bar with a magnifying glass icon and a text input field containing '0.0.0.0/0' with a close button 'X'. To the right of the search bar is an empty text input field. On the far right is a button labeled 'Delete'.

- Click on “Save rules”

Note: This is not the most secure way to give access to the entire world for our cluster. But for our demo purposes it's fine.





# Connection to the EMR master node

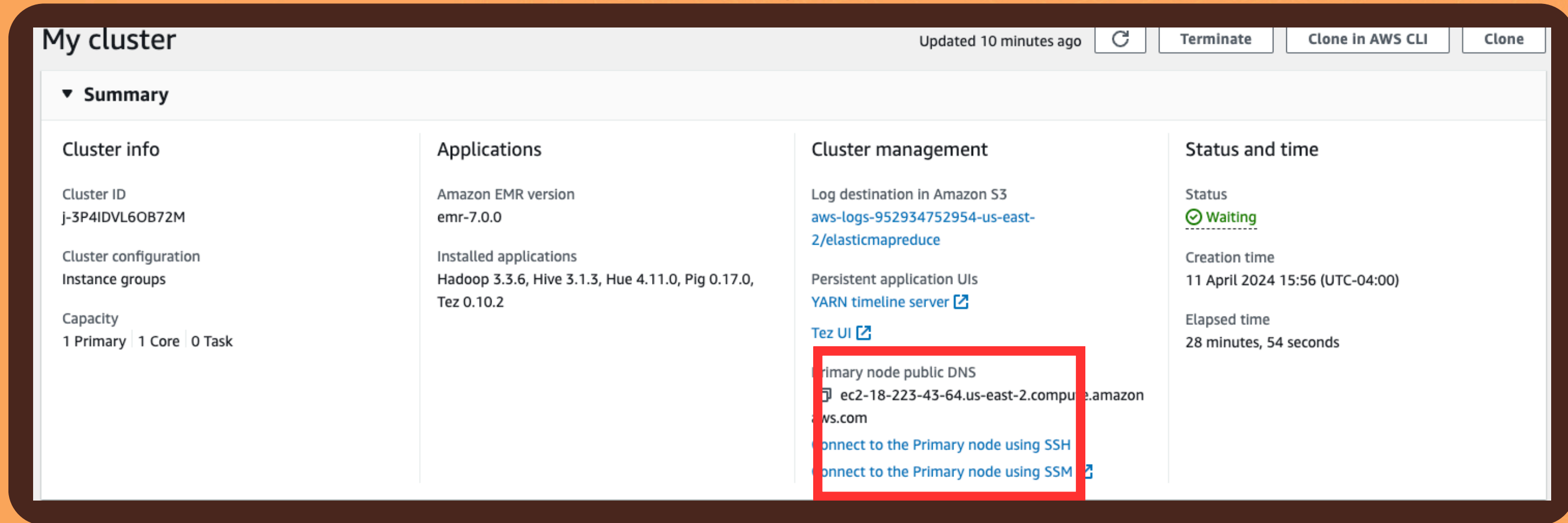




3. Now connect to the EMR master node via putty to perform Hive table creation and ingestion of data

a. Search “EMR” -> Click on Clusters Click on “View Cluster Details”. You will be able to see “Summary Information”

b. Click on “Connect to Master Node using SSH”, there will be your own link.



**My cluster** Updated 10 minutes ago [Refresh](#) [Terminate](#) [Clone in AWS CLI](#) [Clone](#)

▼ **Summary**

| Cluster info                                    | Applications                                                                                  | Cluster management                                                                                                                                                                                                   | Status and time                                         |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Cluster ID</b><br>j-3P4IDVL6OB72M            | <b>Amazon EMR version</b><br>emr-7.0.0                                                        | <b>Log destination in Amazon S3</b><br><a href="#">aws-logs-952934752954-us-east-2/elasticmapreduce</a>                                                                                                              | <b>Status</b><br><span>Waiting</span>                   |
| <b>Cluster configuration</b><br>Instance groups | <b>Installed applications</b><br>Hadoop 3.3.6, Hive 3.1.3, Hue 4.11.0, Pig 0.17.0, Tez 0.10.2 | <b>Persistent application UIs</b><br><a href="#">YARN timeline server</a><br><a href="#">Tez UI</a>                                                                                                                  | <b>Creation time</b><br>11 April 2024 15:56 (UTC-04:00) |
| <b>Capacity</b><br>1 Primary   1 Core   0 Task  |                                                                                               | <b>Primary node public DNS</b><br><a href="#">ec2-18-223-43-64.us-east-2.compute.amazonaws.com</a><br><a href="#">Connect to the Primary node using SSH</a><br><a href="#">Connect to the Primary node using SSM</a> | <b>Elapsed time</b><br>28 minutes, 54 seconds           |

Depending on whether you use Windows or Mac; you should follow the steps listed over here.

Since I am using Mac to connect; I'll follow the steps listed for Mac

c. Open a terminal window. On Mac OS X, choose Applications > Utilities > Terminal. On other Linux distributions, the terminal is typically found at Applications > Accessories > Terminal.

d. To establish a connection to the primary node, enter the following command. Replace `~/CS673_AWS_Tutorial.pem` with the location and filename of the private key file (.pem) that you used to launch the cluster.



If you plan to use an SSH client on a macOS or Linux computer to connect to your Linux instance, use the following command to set the permissions of your private key file so that only you can read it. “`chmod 400 your-key-pair-name.pem`”

```
base) invictus@MacBook-Pro-51 ~ % cd Desktop
base) invictus@MacBook-Pro-51 Desktop % chmod 0400 CS673_AWS_Tutorial.pem
base) invictus@MacBook-Pro-51 Desktop % ssh -i ~/Desktop/CS673_AWS_Tutorial.pem hadoop@ec2-3-141-199-44.us-east-2.compute.amazonaws.com

newer release of "Amazon Linux" is available.
Version 2023.4.20240319:
Version 2023.4.20240401:
run "/usr/bin/dnf check-release-update" for full release and version update info

#_
~\_#####_ Amazon Linux 2023
~~\_#####\
~~\_###|
~~\_#/____ https://aws.amazon.com/linux/amazon-linux-2023
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EEEEEEEEEEEEEEEEEEEE MMMMMMMM MMMMMMMM RRRRRRRRRRRRRRRR
: :::::EE M: :::::M M: :::::M R: :::::R
E: :::::EEEEEEEEEE M: :::::M M: :::::M R: :::::RRRRRRR: :::::R
E: :::::E EEEEE M: :::::M M: :::::M RR: :::::R R: :::::R
E: :::::E M: :::::M: M: M: M: M: R: :::::R R: :::::R
E: :::::EEEEEEEEEE M: :::::M M: M: M: M: R: :::::RRRRRRR: :::::R
E: :::::EEEEEEEEEE M: :::::M M: M: M: M: R: :::::RRRRRRR: :::::R
E: :::::E M: :::::M M: M: M: M: R: :::::R R: :::::R
E: :::::E EEEEE M: :::::M MMM M: :::::M R: :::::R R: :::::R
: :::::EEEEEEEEEE: :::::E M: :::::M M: :::::M R: :::::R R: :::::R
: :::::EEEEEEEEEE: :::::E M: :::::M M: :::::M RR: :::::R R: :::::R
EEEEEEEEEEEEEEEEEEEE MMMMMMMM MMMMMMMM RRRRRRRR RRRRRR
```





1. Create an hdfs directory and grant permission to it

a. Type the below commands on your master node:

***hdfs dfs -mkdir -p /user/hive/warehouse***

***hdfs dfs -chmod g+w /user/hive/warehouse***

2. Extract the features.zip file and unzip it

a. Run the below command: **wget**

***<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/samples/features.zip>***

Note: Ensure you are under /home/hadoop directory and make sure the file features.zip resides in this directory

```
[[hadoop@ip-172-31-0-18 ~]$ hdfs dfs -mkdir -p /user/hive/warehouse
[[hadoop@ip-172-31-0-18 ~]$ b.  hdfs dfs -chmod g+w /user/hive/warehouse
-bash: b.: command not found
[[hadoop@ip-172-31-0-18 ~]$ hdfs dfs -chmod g+w /user/hive/warehouse
[[hadoop@ip-172-31-0-18 ~]$ wget https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/samples/features.zip
--2024-04-11 19:19:37-- https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/samples/features.zip
Resolving docs.aws.amazon.com (docs.aws.amazon.com)... 18.238.25.19, 18.238.25.69, 18.238.25.28, ...
Connecting to docs.aws.amazon.com (docs.aws.amazon.com)|18.238.25.19|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14307 (14K) [application/zip]
Saving to: 'features.zip'

features.zip          100%[=====>]  13.97K  --.-KB/s   in 0s

2024-04-11 19:19:37 (266 MB/s) - 'features.zip' saved [14307/14307]
```



Run command: *unzip features.zip*

```
[hadoop@ip-172-31-0-18 ~]$ unzip features.zip
Archive:  features.zip
  inflating: features.txt
[hadoop@ip-172-31-0-18 ~]$ hive
Hive Session ID = 42248bd4-86a0-4c9d-af92-da3bbbeb107bc
```

3. Create a Hive Table and ingest data into it:
  - a. Type “**Hive**” and hit enter

```
[hadoop@ip-172-31-0-18 ~]$ hive
Hive Session ID = 42248bd4-86a0-4c9d-af92-da3bbbeb107bc

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: false
[hive> CREATE TABLE hive_features (feature_id BIGINT,feature_name STRING ,feature_class STRING,state_alpha STRING, prim_lat_dec DOUBLE, prim_long_dec
DOUBLE ,elev_in_ft BIGINT) ROW FORMAT DELIMITED FIELDS TERMINATED BY '|' LINES TERMINATED BY '\n';
OK
Time taken: 0.571 seconds
[hive> LOAD DATA LOCAL INPATH './features.txt' OVERWRITE INTO TABLE  hive_features;
Loading data to table default.hive_features
OK
Time taken: 0.698 seconds
[hive> SELECT state_alpha, COUNT(*) FROM hive_features GROUP BY state_alpha;
```



b. Create a table named “hive\_features” using the below DDL:

***CREATE TABLE hive\_features (feature\_id BIGINT,feature\_name STRING ,feature\_class STRING,state\_alpha STRING, prim\_lat\_dec DOUBLE, prim\_long\_dec DOUBLE ,elev\_in\_ft BIGINT) ROW FORMAT DELIMITED FIELDS TERMINATED BY '|' LINES TERMINATED BY '\n';***

c. Load the “features.txt” data within this table

***LOAD DATA LOCAL INPATH './features.txt' OVERWRITE INTO TABLE hive\_features;***

e. Run the below query to display a list of states and number of geographic features in each of these states:

***SELECT state\_alpha, COUNT(\*) FROM hive\_features GROUP BY state\_alpha;***

```
[[hadoop@ip-172-31-0-18 ~]$ hive
Hive Session ID = 42248bd4-86a0-4c9d-af92-da3bbeb107bc

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: false
[hive> CREATE TABLE hive_features (feature_id BIGINT,feature_name STRING ,feature_class STRING,state_alpha STRING, prim_lat_dec DOUBLE, prim_long_dec
DOUBLE ,elev_in_ft BIGINT) ROW FORMAT DELIMITED FIELDS TERMINATED BY '|' LINES TERMINATED BY '\n';
OK
Time taken: 0.571 seconds
[hive> LOAD DATA LOCAL INPATH './features.txt' OVERWRITE INTO TABLE  hive_features;
Loading data to table default.hive_features
OK
Time taken: 0.698 seconds
[hive> SELECT state_alpha, COUNT(*) FROM hive_features GROUP BY state_alpha;
```



```
Query ID = hadoop_20240411192335_0e46e2ab-f92b-4263-aa87-50f54cff29ba
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1712862469954_0001)
```

| VERTICES        | MODE      | STATUS    | TOTAL | COMPLETED | RUNNING | PENDING | FAILED | KILLED |
|-----------------|-----------|-----------|-------|-----------|---------|---------|--------|--------|
| Map 1 .....     | container | SUCCEEDED | 1     | 1         | 0       | 0       | 0      | 0      |
| Reducer 2 ..... | container | SUCCEEDED | 2     | 2         | 0       | 0       | 0      | 0      |

```
VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 5.07 s
```

```
OK
AK    20
CA    47
CO    16
CT     4
HI     3
ID    17
IL     5
ME     7
MI    12
MS     2
NC     8
ND     1
NJ     3
NV     5
OH     4
OK     5
PA    16
RI     2
```

Upload readable screenshots of the all the steps of the Assignments as ONE single Word document.

Make sure you terminate the EMR cluster once the assignment is finished else the EC2 instances would run at the backend increasing the cost





Let's rock it!